SPH4U Unit 1 Kinematics - Version 1 Answer Key and Marking Guide

Companion to the 60-mark student paper. Accept equivalent correct methods, directions, and sensible rounding. Difficulty tags below are for planning, not student grading.

A. Multiple Choice (15 marks)

Q	Answer	Difficulty	Key reason
1	A	Easy	Zero net displacement gives zero average velocity.
2	B	Easy	Tangent slope on x-t is instantaneous velocity.
3	C	Easy	Signed area on v-t is displacement.
4	B	Easy	72 ÷ 3.6 = 20 m/s.
5	C	Easy	Gravity remains downward at the apex.
6	C	Medium	Average velocity is (+4 + -2)/2 = +1 m/s; displacement = +3 m.
7	A	Medium	$\sqrt{(6^2 + 8^2)} = 10$ m; $\tan^{-1}(8/6) = 53°$ N of E.
8	A	Medium	$\Delta v = v_f - v_i$ = (-4 E + 3 N) m/s.
9	C	Medium	Horizontal velocity is constant: 12 × 2.5 = 30 m.
10	B	Medium	$\sqrt{(3^2 + 4^2)} = 5.0$ m/s; $\tan^{-1}(3/4) = 37°$ E of N.
11	C	Hard	$0 = 12 + 5t - 4.9t^2$ gives t = 2.16 s.
12	B	Hard	Positive areas: 6 + 4.5 m; negative area magnitude: 0.5 m. Total distance = 11 m.
13	A	Hard	$v_{plane/air}$ = (-20 E + 60 N) km/h = 63.2 km/h [18.4° W of N].
14	C	Hard	a_{avg} = [(10 E - 10 N) m/s]/2 s = (5 E - 5 N) m/s².
15	B	Hard	Relative speed = 36 km/h = 10 m/s; relative distance = 200 + 4 = 204 m; t = 20.4 s.

B. True or False (5 marks)

Q	Answer	Difficulty	Key reason
16	T	Easy	Only the vertical component is zero at the apex.
17	T	Easy	This is the definition of average speed.
18	F	Medium	A negative acceleration can increase speed if velocity is also negative.
19	T	Medium	Vertical motion determines landing time; horizontal launch speed does not.
20	F	Hard	A finite northward speed reduces drift but cannot cancel eastward current; aim west of north.

C. Lab Analysis (10 marks)

21a (2 marks, Easy). Δx = 19.2 - 0.0 = **19.2 m [E]**. Award 1 for endpoint subtraction and 1 for value/direction.

21b (2 marks, Easy). $v_{avg} = \Delta x/\Delta t$ = 19.2/4.0 = **4.8 m/s [E]**. Award 1 for method and 1 for value/unit/direction.

21c (2 marks, Medium). Plot the five coordinates (0,0), (1,1.2), (2,4.8), (3,10.8), (4,19.2); label t (s) and x (m [E]); draw a smooth upward-curving trend. Award 1 for points/scale and 1 for labelled axes/curve.

21d (2 marks, Medium). Symmetric secant about 2.0 s: $(x_3 - x_1)/(3.0 - 1.0) = (10.8 - 1.2)/2.0 =$ **4.8 m/s [E]**. For this quadratic data, the symmetric secant equals the instantaneous velocity at the midpoint. Award 1 for method and 1 for result/explanation.

21e (2 marks, Medium). At t = 2.0 s, $4.8 = \frac{1}{2}a(2.0)^2$, so **$a = 2.4$ m/s² [E]**. The increasing slope/curvature of the x-t graph shows non-uniform velocity. Award 1 for acceleration and 1 for graph interpretation.

D. Applications (30 marks)

22a (2 marks, Easy). Total distance = 120 + 50 = **170 m**.

22b (3 marks, Easy). Δr = (120 E + 50 N) m. Magnitude = $\sqrt{(120^2 + 50^2)}$ = **130 m**; $\theta = \tan^{-1}(50/120)$ = **22.6° N of E**. Award 1 for components/sketch, 1 for magnitude, 1 for direction.

22c (2 marks, Easy). Average speed = 170/40 = **4.25 m/s**. Award 1 for expression and 1 for result.

22d (3 marks, Medium). Average velocity = $\Delta r/\Delta t$ = 130/40 = **3.25 m/s [22.6° N of E]**. Its magnitude is smaller than average speed because displacement (130 m) is shorter than distance (170 m). Award 1 for calculation, 1 for direction, 1 for explanation.

23a (2 marks, Easy). $v_{ix} = 18.0 \cos 30.0°$ = **15.6 m/s [E]**; $v_{iy} = 18.0 \sin 30.0°$ = **9.00 m/s [up]**. One mark each.

23b (2 marks, Medium). With launch height y_0 = 20.0 m, $0 = 20.0 + 9.00t - 4.90t^2$. The positive root is **t = 3.14 s**. Award 1 for vertical equation and 1 for correct root.

23c (2 marks, Medium). $\Delta x = v_{ix} t = (15.588...)(3.1376...)$ = **48.9 m [E]**. Award 1 for method and 1 for result.

23d (4 marks, Hard). v_{fx} = **15.6 m/s [E]**, $v_{fy} = 9.00 - 9.80(3.1376...)$ = **-21.7 m/s**. Magnitude = $\sqrt{(15.588^2 + 21.749^2)}$ = **26.8 m/s**. Direction = $\tan^{-1}(21.749/15.588)$ = **54.4° below the eastward horizontal**. Award 1 for each component, 1 for magnitude, 1 for direction.

24a (2 marks, Hard). Vertical motion begins with $v_{iy} = 0$: $45.0 = \frac{1}{2}(9.80)t^2$, so **t = 3.03 s**. Award 1 for equation and 1 for time.

24b (2 marks, Hard). $v_{package/cart} = v_{package/ground} - v_{cart/ground}$ = (20.0 E + 12.0 N) - (5.0 E) = **(15.0 E + 12.0 N) m/s**. Award 1 for frame relation and 1 for components.

24c (3 marks, Hard). In 3.03046 s, package displacement is (60.61 E + 36.37 N) m. Cart displacement is (15.15 E + 0 N) m. Thus the cart must start at **(45.5 m E, 36.4 m N)** relative to P. Award 1 for package position, 1 for cart motion, 1 for starting coordinates. Equivalent use of the relative velocity from part b earns full credit.

24d (2 marks, Hard). Initial separation = $\sqrt{(45.4569^2 + 36.3655^2)}$ = **58.2 m [38.7° N of E]**. Award 1 for magnitude and 1 for bearing.

24e (1 mark, Hard). Immediately after release, the package keeps the drone's horizontal ground velocity: **(20.0 E + 12.0 N) m/s**. Its horizontal velocity does not jump to zero because there is no horizontal impulse at release.

Difficulty audit

Section	Easy	Medium	Hard	Total
A: Multiple Choice	5	5	5	15
B: True or False	2	2	1	5
C: Lab Analysis	4	6	0	10
D: Applications	9	7	14	30
Total	**20**	**20**	**20**	**60**

The source training pages are harder than some items in this first practice version. Their multi-step vector, projectile, and relative-motion style is represented especially in questions 11-15, 23d, and 24.